(ADVANCED COUNTING TECHNIQUES)

8.1 - APPLICATIONS OF RECURRENCE RELATIONS

RECURRENCE RELATIONS

A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more previous terms of the sequence, namely $a_0, a_1, a_2, \cdots, a_{n-1}$ for all integers n with $n \geq n_0$, where n_0 is non-negative integer.

A sequence is called a solution of recurrence relation if its terms satify the recurrence relation.

The initial conditions for sequence specify the terms that precede the first term where recurrence relation takes effect.

~~Example #1~~

MODELING WITH RECURRENCE RELATIONS

We can use recurrence relations to model a wide variety of problems such as:

- Counting Rabbits on an island
- Determining the number of moves in Tower of Hanoi Puzzle
- Counting bit strings with certain properties

EXAMPLE # 1

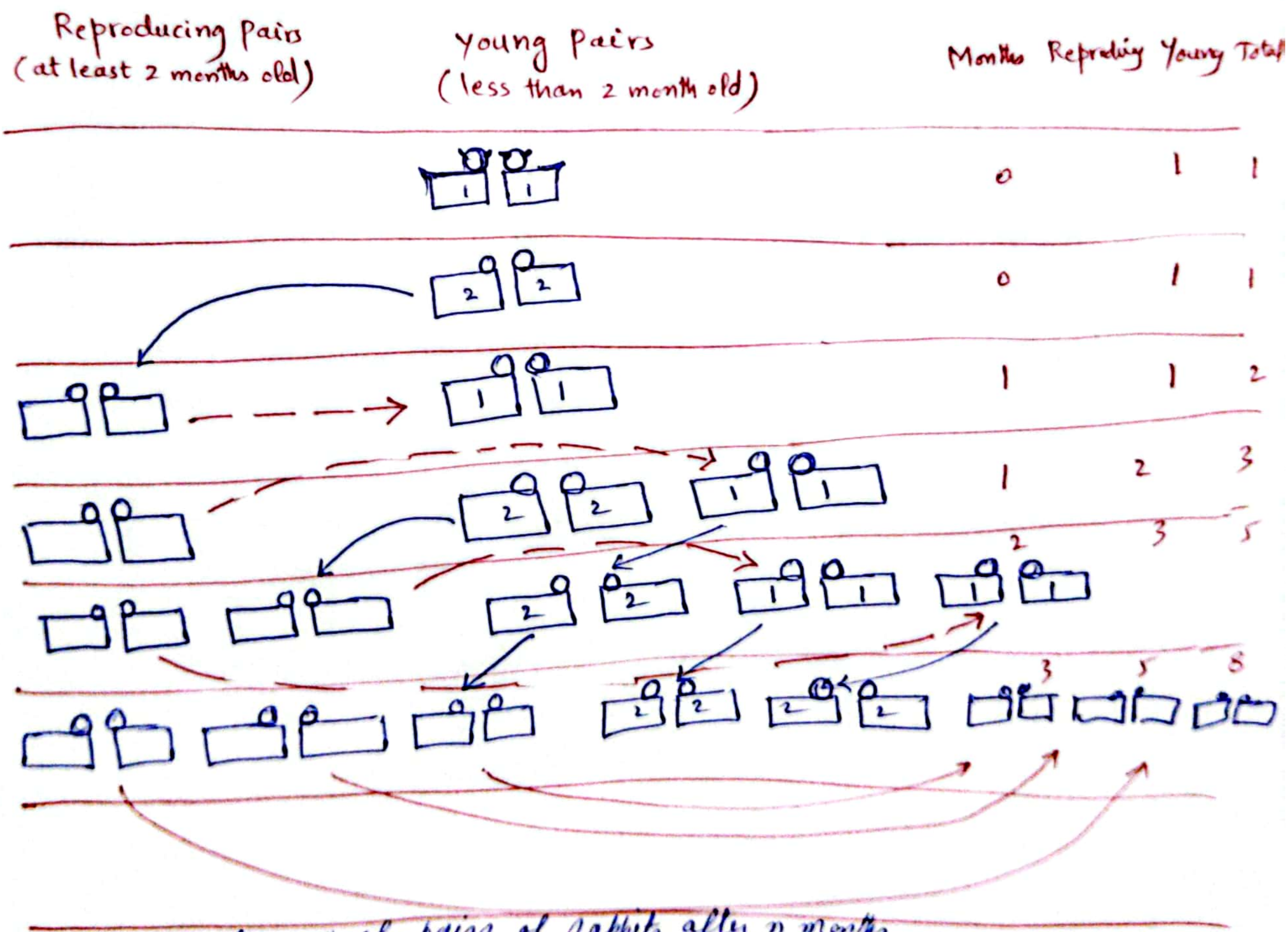
Rabbits and Fibonacci Numbers :

A young pair of rabbits (one of each sex) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month.

Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that no rabbits ever die.

Sol :-

This problem is shown by the following figure:



Let f_n = # of pairs of rabbits after n months

so, $f(n)$ is $f_n = f_{n-1} + f_{n-2}$ where $f_1 = 1$, $f_2 = 2$

EXAMPLE #2

The Tower of Hanoi :

A popular puzzle of the late nineteenth century invented by the French mathematician Edouard Lucas, called Tower of Hanoi, consists of 3 pegs mounted on a board together with disks of different sizes. Initially these disks are placed on peg 1 in order of size, with largest on the bottom. The rules of the puzzle allow disks to be moved one at a time from one peg to another as long as disk is never placed on top of a smaller disk.

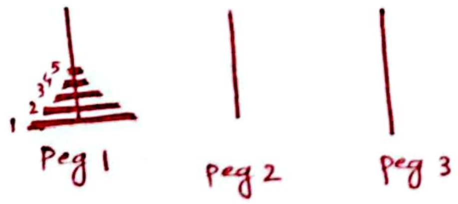
The goal of the puzzle is to have all disks on the second peg in order of size, with the largest on the bottom.

Let H_n, denote the number of moves needed to solve the Tower of Hanoi problem with n disks.

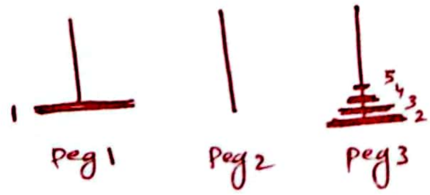
Set up a recurrence relation for the sequence $\{H_n\}$.

Sol :

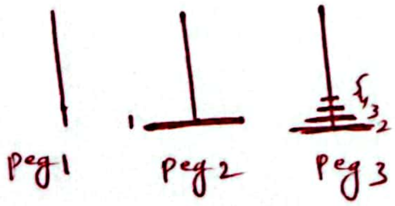
Begin with n disks on Peg 1.



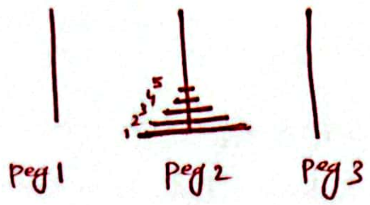
1. We can transfer the top $n-1$ disks following the rules of the puzzle to peg 3 using H_{n-1} moves. We keep the largest disk fixed using these moves.



2. We use one move to transfer the largest disk to peg 2.



3- Finally, we transfer the $n-1$ disks on peg 3 to peg 2 using H_{n-1} moves, placing them on top of the largest disk.



Total number of moves needed to solve the Tower of Hanoi problem with n disks:

$$H_n = H_{n-1} + 1 + H_{n-1}$$

$$\boxed{\begin{array}{l} H_n = 2H_{n-1} + 1 \\ H_1 = 1 \end{array}}$$

recurrence relation

We can solve it as:

1 $\quad H_n = 2\underline{H_{n-1}} + 1$

2 $\quad = 2(2H_{n-2}+1)+1 = 2^2\underline{H_{n-2}} + 2 + 1$

3 $\quad = 2^2(2H_{n-3}+1) + 2 + 1 = 2^3 H_{n-3} + 2^2 + 2 + 1$

$\vdots$

k $\quad = 2^k H_{n-k} + 2^{k-1} + \cdots + 2^2 + 2 + 1$

$$= 2^{n-1} H_{n-(n-1)} + 2^{n-1-1} + \cdots + 2^2 + 2 + 1 \qquad \text{put } n-k=1,\ k=n-1$$

$$= 2^{n-1} H_1 + 2^{n-2} + \cdots + 2^2 + 2 + 1$$

$$= \underbrace{2^{n-1}\cdot 1 + 2^{n-2} + \cdots + 2^2 + 2 + 1}$$

$$H_n = 2^n - 1$$

MYTH:

Monks are transferring 64 gold disks from 1 peg to another, according to rules. Myth says that world will end when they finish the puzzle.

$H_{64} = 2^{64} - 1$ = 500 billion years to complete transfer assuming 1 move will take 1 sec.

EXAMPLE#3:

Find a recurrence relation and give initial conditions for the number of bit strings of length n that donot have two Consecutive 0s. How many such bit strings are there of length five?

Sol:

The bit strings of length n can be divided into

- those that end in 1, and
- those that end in 0

	1
	0

we assume that

a_n = # of bit strings of length n that donot have 2 Consecutive 0s

We can write a_n as:

a_n = # of strings of length n, end with 1 that donot have 2 Consecutive 0s +
# of strings of length n, end with 0 that donot have 2 Consecutive 0s.

FIRST: Look at the bit strings ending with 1 : $\underbrace{a_{n-1} \text{ bit strings}}_{n-1}$ | 1

Use recursive definition to Count bit strings.

Do we have any restriction on strings of length $n-1$ that donot have Consecutive 0s (i.e the digit left before 1). The answer is No.
It should be general.

So, we can say that

a_{n-1} be the Count of bit strings of length ~~a_{n-1}~~ $n-1$, that donot have 2 Consecutive 0s.

SECOND: Look at the bit strings end with 0 :

	0

Do we have any restriction on strings of length $n-1$ that donot have Consecutive 0s (i.e the digit left before 0). The answer is YES.

$(n-2)$		
No string	0	0
a_{n-2} bit strings	1	0

a_{n-2} be the # of strings of length $n-2$ that donot have 2 Consecutive 0s.

So, Recurrence relation becomes

a_{n-1}			1
zero		0	0
a_{n-2}		1	0

$$a_n = a_{n-1} + 0 + a_{n-2}$$

$$a_n = a_{n-1} + a_{n-2}$$

Initial Conditions: $a_1 = 2$ ($n=1$, two possible strings 0, 1)
$a_2 = 3$ ($n=2$, three possible strings 01, 10, 11)

Recurrence Relation :

$$a_n = a_{n-1} + a_{n-2} \quad , \; a_1 = 2 \quad , \; a_2 = 3$$

Same as Fibonaci Sequence but with different initial Conditions.

⊙ How many Such bit strings of length five ?

To obtain a_5 , we use recurrence relation as:

$$a_n = a_{n-1} + a_{n-2} \qquad a_1 = 2, a_2 = 3$$

$$a_3 = a_2 + a_1$$
$$= 3 + 2 = 5$$

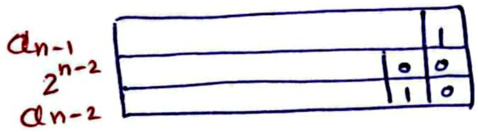
$$a_4 = a_3 + a_2$$
$$= 5 + 3 = 8$$

$$a_5 = a_4 + a_3$$
$$= 8 + 5 = 13$$

0	0 0 0 0 0
1	0 0 0 0 1
2	0 0 0 1 0
⋮	
31	1 1 1 1 1

13 bit strings of length 5 that ~~have~~ donot have Consecutive 0s.

Example :- # of bit strings that have 2 Consecutive 0s .



n=1
0 ✓
1 ✓ (no one)

n=2
00 ✓
01
10
11 ✗

n= 3
000 ✓
001 ✓
010
011
100 ✓
101
110
111

$$a_n = a_{n-1} + 2^{n-2} + a_{n-2} \quad , \; a_1 = 0 \; , \; a_2 = 1$$

$$a_3 = a_2 + 2^{3-2} + a_{3-2} = 1 + 2^1 + 0 = 3$$

$$a_4 = a_3 + 2^2 + a_2 = 3 + 4 + 1 = 8$$

$$a_5 = a_4 + 2^3 + a_3 = 8 + 8 + 3 = 19$$

19 bit strings of length 5 that have 2 Consecutive 0's .

EXAMPLE #4

A Computer system Considers a string of decimal digits a valid Codeword if it Contains an even number of 0 digits. For instance, 1230407869 is valid, whereas 120987045608 is not valid. Let a_n be the number of valid n-digit Codeword. Find a recurrence relation for a_n.

Sol:-

a_n = # of valid n-digit Codewords
(# of strings of decimal digits of length n with even # of 0 digits)

There are two ways to form valid strings with n digits with one fewer digit.

FIRST WAY : 0 (string of length n−1 with odd # of 0 digits)
SECOND WAY : $\underbrace{\text{(digit other than 0)}}_{1-9}$ (string of length n−1 with even # of 0 digits)

First Way

0	odd # of 0 digits
1	even # of 0 digits
2	" " "
3	" " "
4	" " "
⋮	" " "
9	" " "

Second Way

total # of Codewords = Sum Rule

FIRST CASE : 0 (string of length (n−1) with odd # of 0 digits)

0	$\overbrace{\qquad\qquad}^{n-1}$

1. There are total 10^{n-1} strings of decimal digits of length n−1 (including valid Codewords).

2. There are a_{n-1} valid Code words (i.e., strings of decimal digits with even # of 0)

As leftmost digit is 0, we need strings of decimal digits with odd # of 0 digits in its right.

$\Rightarrow$ # of strings of length n−1 with odd # of 0 digits $= 10^{n-1} - a_{n-1}$

1 + (n−1)	
0	odd # of strings

So There are
$(10^{n-1} - a_{n-1})$ ——— Ⓐ
strings of decimal digits of length n with even # of 0 digits.

SECOND CASE: (1 − 9)(string of Length n−1 with even # of 0 digits)

1	Even # of 0 digits
2	" " "
3	" " "
4	" " "
⋮	
9	" " "

Do we have any restriction on strings of length n−1. The answere is No. It should be general.

i.e.,

There are a_{n-1} total strings of decimal digits of length n−1 with even # of 0.

$\Rightarrow$

1. a_{n-1} Count for strings starting with 1
2. a_{n-1} Count for strings starting with 2
3. a_{n-1} Count for strings starting with 3

⋮

9. a_{n-1} Count for strings starting with 9

So, There are

$$(\overset{①}{a_{n-1}} + \overset{②}{a_{n-1}} + \overset{③}{a_{n-1}} + \cdots + \overset{⑨}{a_{n-1}}) = 9\,a_{n-1}$$

total strings with even # of 0 digits of length n.

1 + (n−1)

1	

All valid strings of length n are:

$$a_n = \text{First Case Count} + \text{Second Case Count}$$

$$= 10^{n-1} - a_{n-1} + 9\,a_{n-1}$$

$$a_n = 8\,a_{n-1} + 10^{n-1}$$

$$a_1 = 9$$

$$a_2 = 8\,a_{2-1} + 10^{2-1}$$

$$= 8\,a_1 + 10$$

$$a_2 = 8 \times 9 + 10 = 82$$

a_2

— —

00 1

11 12 ⋮ 19 9

21 ⋮ 29 9

31 ⋮ 39 9

41 ⋮ 49 9

51 ⋮ 59 9

61 ⋮ 69 9

71 ⋮ 79 9

81 ⋮ 89 9

91 ⋮ 99 9

81 + 1 = 82

$a_1 = 9$

1
2
3
4
5
6
7
8
9

(valid)

EXAMPLE # 5

Find a recurrence relation for C_n, the number of ways to parenthesize the product of $n+1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \cdots \cdot x_n$, to specify the order of multiplication. For example, $C_3 = 5$ because there are five ways to parenthesize $x_0 \cdot x_1 \cdot x_2 \cdot x_3$ to determine the order of multiplication:

1. $((x_0 \cdot x_1) \cdot x_2) \cdot x_3$ 2. $(x_0 \cdot (x_1 \cdot x_2)) \cdot x_3$ 3. $(x_0 \cdot x_1) \cdot (x_2 \cdot x_3)$
4. $x_0 \cdot ((x_1 \cdot x_2) \cdot x_3)$ 5. $x_0 \cdot (x_1 \cdot (x_2 \cdot x_3))$

Sol:

What are different parenthesization we can carry out the following multiplications:

1. $x_0 \cdot x_1$: in case of two numbers

   Possible # of ways to parenthesize $= (x_0 \cdot x_1)$

   i.e $C_1 = 1$ (one way)

2. $x_0 \cdot x_1 \cdot x_2$: in case of three numbers ($n+1$ numbers)

   possible parenthesization = 1. $(x_0 \cdot x_1) \cdot x_2$ 2. $x_0 \cdot (x_1 \cdot x_2)$

   i.e $C_2 = 2$

3. $x_0 \cdot x_1 \cdot x_2 \cdot x_3$: in case of four numbers ($n+1$ numbers)

   Possible parenthesization = $\underbrace{x_0 \cdot [x_1 \cdot x_2 \cdot x_3]}_{2}$, $\underbrace{(x_0 \cdot x_1) \cdot (x_2 \cdot x_3)}_{1}$, $\underbrace{[x_0 x_1 x_2] \cdot x_3}_{2}$

   i.e $C_3 = 5$

WHY PARENTHESIZATION IS IMPORT

$A = 100 \times 2 \quad B = 2 \times 50 \quad C = 50 \times 3$

$(AB)C = [A]_{100\times 2} [B]_{2\times 50} = [AB]_{100 \times 50}$
$= 100 \times 2 \times 50$
$= 10{,}000$

$[AB]_{100\times 50} [C]_{50\times 3} = 100 \times 50 \times 3 = 15000$

$(AB)C = 10{,}000 + 15000 = \boxed{25000}$

$A(BC) = 600 + 300 = 900$ multiplicat

Assume that

$$\underbrace{(x_0 \cdot x_1 \cdot x_2 \cdots x_K)}_{K+1 \text{ numbers}} \cdot \underbrace{(x_{K+1} \cdots x_n)}_{n-K \text{ numbers}}$$

Possible ways $\quad C_K \cdot C_{n-K-1}$

This is only for one K. If you want to do all parenthesization, we need to consider all values of K starting from 0 upto $n-1$.

$$\left.\begin{aligned} C_n &= \sum_{K=0}^{n-1} C_K \cdot C_{n-K-1} \\ C_0 &= 1 \\ C_1 &= 1 \end{aligned}\right\} \text{Catalan Numbers}$$

2 options $(x_0)(x_1 x_2 x_3)$ $K=0$
$(x_0 x_1)(x_2 x_3)$ $K=1$
2 options $(x_0 x_1 x_2)(x_3)$ $K=2$

$C_3 = C_0 C_{3-0-1} + C_1 C_{3-1-1} + C_2 C_{3-2-1}$
$= C_0 C_2 + C_1 C_1 + C_2 C_0$
$= 1 \cdot 2 + 1 \cdot 1 + 2 \cdot 1$
$= 5$

$(x_0)(x_1 x_2 x_3)$
① $= x_0 (x_1 x_2) x_3$
② $= (x_0)(x_1 \cdot (x_2 x_3))$